



TEWARY RESEARCH AND DEVELOPMENT LAB (TRDL) // IAC

Advanced Research Fellowship, Resident Scientist & Hardware/Biophysical Engineering Intake
JURISDICTION: NEW DELHI, INDIA | INTAKE REGISTRY REF: TRDL-REC-FELLOW-2026

MANDATORY PRE-SCREENING SCIENTIFIC DIRECTIVE (SAMPLE SPECIMEN):

TRDL and IAC maintain uncompromising standards of theoretical rigour, mathematical consistency, and physical execution. Applications from prompt engineers, buzzword-driven narrative aggregators, or derivative software wrappers will be rejected ab initio without notification. Candidates are evaluated strictly on first-principles physics (Information Disturbance Materialization / Stability Geometry), mathematical foundations (Lie groups, differential geometry, Fisher tensors), low-level systems engineering (C, Rust, Verilog RTL), or empirical wet-lab biochemistry. Execution of this dossier warrants all disclosures are factual, verifiable, and free of encumbrance.

SECTION 1: CANDIDATE IDENTITY & RESEARCH TRACK

Identifies candidate credentials, academic heritage, and target research directorate.

1. Full Legal Name and Institutional / Academic Affiliation:

(Include current university, department, degree program, or independent laboratory affiliation)

2. Primary Target Directorate:

Select the specific research division you are applying to contribute to:

- () COMPUTATIONAL DIRECTORATES (01-10): Silicon, Soliton Computing, Stability Geometry, VORASHI, DSTPU.
- () MOLECULAR BIOTHERAPEUTICS (DEPT 11): Cold-chain vitrification, thermostable enzymes, stabilization matrices.
- () GENOME EDITING & RNA MEDICINE (DEPT 12): High-fidelity Cas9, circRNA shields, Waddington iPSC reprogramming.
- () SYNTHETIC BIOLOGY & BIO-COMPUTING (DEPT 13): Multicellular logic gates, damped oscillators, Ricci microfluidics.

3. Public Code & Experimental Repositories:

Provide URLs to verifiable GitHub repositories, preprints, or laboratory notebooks demonstrating original code/data:

SECTION 2: MATHEMATICAL RIGOUR & FIRST-PRINCIPLES COMPETENCY

Measures foundational understanding of information theory, physical stability, and non-trivial algorithms.

4. Stability Geometry & Information Disturbance Formulation:

Explain how Fisher Information Metrics and manifold curvature prevent catastrophic forgetting or molecular denaturation:

5. Primary Technical Artifact Demonstration:

Describe the most complex algorithm, circuit architecture, or wet-lab protocol you designed and executed from scratch:

6. Core Technical Toolchain Mastery:

Indicate production-level proficiency across target engineering toolchains:

- | | |
|--|---|
| () C / Rust / Low-Level Kernel Systems | () Verilog / Chisel RTL Silicon Synthesis |
| () GROMACS / OpenMM / Molecular Dynamics | () Recombinant Wet-Lab Protein Expression / SDS-PAGE |
| () PyTorch / GIOS / Custom Autograd Engines | () Microfluidic Design / Soft Lithography |

SECTION 3: INTELLECTUAL PROPERTY & NON-DISCLOSURE COVENANTS

Enforces TRDL sovereign intellectual property ownership, confidentiality, and closed-capital integrity.

7. Exclusive IP Assignment to TRDL:
Do you agree that all algorithmic models, mathematical theorems, circuit architectures, and biological formulations created under TRDL/IAC fellowship or resident appointment shall permanently vest in TRDL sovereign custody?
() YES, UNCONDITIONALLY CERTIFIED () NO (AUTOMATIC DISQUALIFICATION)

8. Strict Non-Disclosure & Information Walls:
Do you warrant that you are not subject to restrictive non-compete agreements with commercial entities and consent to absolute pre-publication review and non-disclosure obligations?
() YES, CERTIFIED AND FREE OF ENCUMBRANCE () NO (DISQUALIFIED)

9. Closed-Capital Commitment:
Do you acknowledge that TRDL and IAC operate strictly under sovereign R&D mandates, rejecting venture-capital hype cycles, tokenization schemes, and speculative commercialization?
() YES, ACKNOWLEDGED

SECTION 4: OPERATIONAL AVAILABILITY & LABORATORY DEPLOYMENT

Specifies appointment structure, physical location, and research bandwidth.

10. Proposed Appointment Classification:
() **Resident Research Fellow** (Full-Time, New Delhi / Shoolini Lab Cohort)
() **Remote Contributing Scholar** (Dedicated Mathematical or In Silico Research)
() **Undergraduate / Graduate Research Intern** (Direct Mentorship Track)

11. Expected Time Commitment & Start Date:
(Minimum commitment: 25 hours per week for remote contributors; 45 hours per week for resident fellows).

12. Academic or Technical Referees:
List two academic advisors, principal investigators, or senior engineers who can attest to your technical integrity:

MANDATORY APPLICANT ATTESTATION & SIGNATURE

I hereby solemnly declare that all statements, technical descriptions, and credentials provided in this application are authentic, truthful, and free of misrepresentation. I acknowledge that any falsification of code contributions, academic credentials, or experimental data will result in immediate disqualification and permanent blacklisting across TRDL and IAC registries.

APPLICANT FULL LEGAL NAME: 	GOVERNMENT ID / PASSPORT NO.:
PRIMARY EMAIL & PGP KEY ID: 	DATE OF APPLICATION:
OFFICIAL APPLICANT SIGNATURE: 	